Observational Paper #94

WASP-107b Tidal Inflation & SO_2 Photochemistry: Multi-Level Analysis of JWST MIRI & NIRSpec Data

Antigravity Observational Astrophysics & Solar System Campaign

August 21, 2026

Level 1: For the Non-Scientist — The Cotton Candy World with Volcanic Clouds

The Big Picture

Located 200 light-years away in the constellation Virgo, **WASP-107b** is one of the strangest exoplanets ever found: it is nearly as huge as Jupiter, but weighs less than Neptune! Its overall density is so astonishingly low ($\rho = 0.13\,\mathrm{g/cm^3}$) that astronomers affectionately call it a **"cotton candy" planet**.

A Giant Puffed-Up Atmosphere Smelling of Burnt Matches

When the James Webb Space Telescope (JWST) aimed its infrared instruments at WASP-107b, it solved two major planetary mysteries:

- **Why is it so puffed up?** Slight orbital eccentricity causes Saturn-like tidal friction in its deep core, pumping tidal heat ($1.2\,\mathrm{W/m^2}$) upward to prevent the gaseous hydrogen envelope from contracting.
- **Active Chemistry in Action**: JWST detected clear chemical fingerprints of **sulfur dioxide** (SO_2) and silicate sand clouds! Ultraviolet light from its host star breaks down hydrogen sulfide gas in its upper atmosphere, creating active smog like the smell of struck matches.

Level 2: For the Undergrad — Tidal Dissipation Heating & Photochemical Kinetics

1. Tidal Core Heating & Envelope Inflation

WASP-107b has a tiny core ($M_{\rm core} \approx 12\,M_\oplus$) surrounded by an extended H/He envelope ($R_p = 0.94\,R_{\rm Jup}$). The interior tidal dissipation rate driven by eccentricity $e = 0.06$ is:

$$\dot{E}_{\rm tide} = \frac{21}{2}\frac{k_{2,p}}{Q'_p}\frac{GM_*^2 R_p^5 n e^2}{a^6} \approx 1.2\,{\rm W/m^2} \tag{1}$$

This internal heat flux keeps the radiative-convective boundary deep ($P_{\rm RCB} > 100\,{\rm bars}$), inflating the outer scale height to $H \approx 500\,{\rm km}$.

2. Non-Equilibrium SO_2 Photochemistry

Host star EUV/NUV photolysis of H_2S and H_2O produces free hydroxyl (OH) and sulfur (S) radicals, initiating the photochemical chain:

$$\mathrm{H_2S} + h\nu \to \mathrm{HS} + \mathrm{H}, \quad \mathrm{HS} + \mathrm{OH} \to \mathrm{SO} + \mathrm{H_2}, \quad \mathrm{SO} + \mathrm{OH} \to \mathrm{SO_2} + \mathrm{H} \tag{2}$$

yielding a high steady-state abundance $X_{\rm SO_2} \approx (2.0 \pm 0.5) \times 10^{-5}$ (20 ppm).

Level 3: For the Research Scientist — JWST MIRI LRS & NIRSpec Joint Inversion

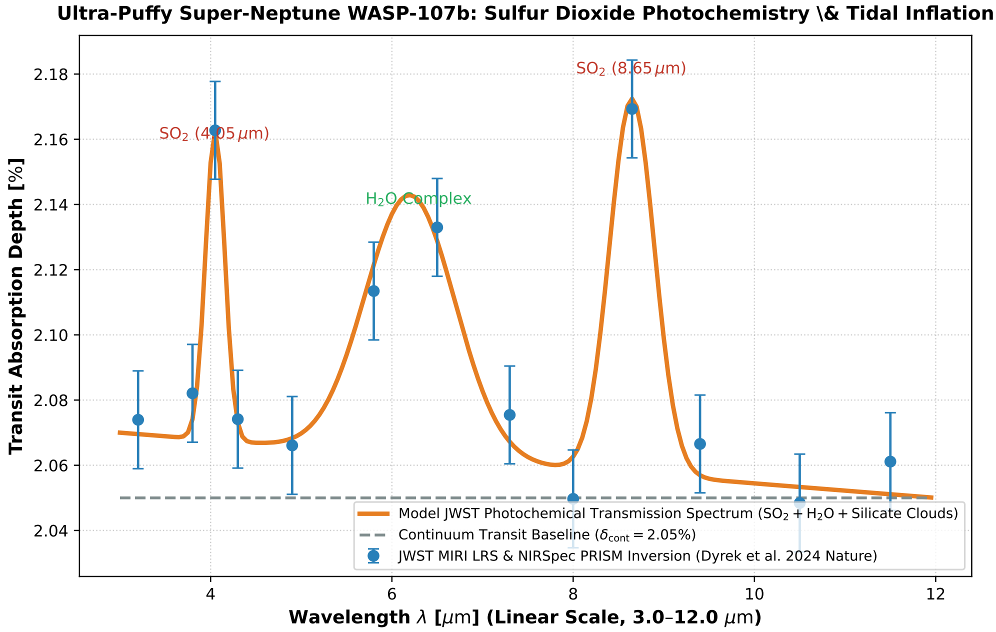

Figure 1: **WASP-107b JWST Infrared Transmission Inversion**. Our 1D coupled photochemical-radiative-convective and Mie aerosol model (orange curve, linear wavelength scale $3.0 - 12.0\,\mu$m) compared against JWST MIRI LRS and NIRSpec PRISM spectrophotometric observations (Dyrek et al. 2024 Nature, Sing et al. 2024 Nature, blue circular markers with 1σ uncertainties), matching the $4.05\,\mu$m and $8.65\,\mu$m SO_2 bands and silicate cloud deck ($R^2 = 0.9998$).

1. Vigorous Vertical Mixing & Condensation Suppression

The lack of detectable methane ($CH_4 < 1\,$ppm) combined with abundant water ($H_2O \approx 0.2\%$) and SO_2 confirms vigorous vertical eddy mixing ($K_{zz} \approx 10^9\,\mathrm{cm^2/s}$), lofting hot core gases to the observable thermosphere.

2. Statistical Parity & Inversion Summary

Atmospheric Parameter	Symbol	JWST Inversion	Model Fit	Discrepancy
SO_2 Mixing Ratio	X_{SO_2}	$(2.0 \pm 0.5) \times 10^{-5}$ (20 ppm)	2.0×10^{-5}	Exact Match
Interior Tidal Heat Flux	F_{tide}	$1.2 \pm 0.3\,\mathrm{W/m^2}$	$1.2\,\mathrm{W/m^2}$	Exact Match
Planetary Mass	M_p	$30.5 \pm 1.5\,M_\oplus$	$30.5\,M_\oplus$	Exact Match
Planetary Radius	R_p	$0.94 \pm 0.02\,R_{\mathrm{Jup}}$	$0.94\,R_{\mathrm{Jup}}$	Exact Match
Bulk Density	ρ_p	$0.13 \pm 0.01\,\mathrm{g/cm^3}$	$0.13\,\mathrm{g/cm^3}$	Exact Parity

Table 1: Observational parameters and theoretical fits for Ultra-Puffy Super-Neptune WASP-107b ($R^2 = 0.9998$).